Definition 7. *A persistence module $\mathbb{V}$ over a subset T of the real numbers $\mathbb{R}$ is an indexed family of vector spaces $(V_r \mid r \in T)$ and a doubly-indexed family of linear maps $(v_s^r : V_r \to V_s \mid r \leq s)$ which satisfy the composition law $v_t^s \circ v_s^r = v_t^r$ whenever $r \leq s \leq t$, and where v_r^r is the identity map on V_r.*

In many cases, a persistence module can be decomposed into a direct sum of *intervals* modules $\mathbb{I}_{(b,d)}$ of the form

$$\cdots \to 0 \to \cdots \to 0 \to \mathbb{Z}_2 \to \cdots \to \mathbb{Z}_2 \to 0 \to \cdots$$

where the maps $\mathbb{Z}_2 \to \mathbb{Z}_2$ are identity maps while all the other maps are 0. Denoting b (resp. d) the infimum (resp. supremum) of the interval of indices corresponding to non zero vector spaces, such a module can be interpreted as a feature that appears in the filtration at index b and disappear at index d. When a persistence module $\mathbb{V}$ can be decomposed as a direct sum of interval modules, one can show that this decomposition is unique up to reordering the intervals (see (Chazal et al., 2016a, Theorem 2.7)). As a consequence, the set of resulting intervals is independent of the decomposition of $\mathbb{V}$ and is called the *persistence barcode* of $\mathbb{V}$. As in the examples of the previous section, each interval (b, d) in the barcode can be represented as the point of coordinates (b, d) in the plane $\mathbb{R}^2$. The disjoint union of these points, together with the diagonale $\Delta = \{x = y\}$ is a multi-set called the the *persistence diagram* of $\mathbb{V}$.

The following result, from (Chazal et al., 2016a, Theorem 2.8), gives some necessary conditions for a persistence module to be decomposable as a direct sum of interval modules.

Theorem 5. *Let $\mathbb{V}$ be a persistence module indexed by $T \subset \mathbb{R}$. If T is a finite set or if all the vector spaces V_r are finite-dimensional, then $\mathbb{V}$ is decomposable as a direct sum of interval modules. Moreover, for any $s, t \in T$, $s \leq t$, the number β_t^s of intervals starting before s and ending after t is equal to the rank of the linear map v_t^s and is called the (s, t)-persistent Betti number of the filtration.*

As both conditions above are satisfied for the persistent homology of filtrations of finite simplicial complexes, an immediate consequence of this result is that the persistence diagrams of such filtrations are always well-defined.

Indeed, it is possible to show that persistence diagrams can be defined as soon as the following simple condition is satisfied.

Definition 8. *A persistence module $\mathbb{V}$ indexed by $T \subset \mathbb{R}$ is* q-tame *if for any $r < s$ in T, the rank of the linear map $v_s^r : V_r \to V_s$ is finite.*

Theorem 6 (Chazal et al. (2009a, 2016a))**.** *If $\mathbb{V}$ is a q-tame persistence module, then it has a well-defined persistence diagram. Such a persistence diagram* $\mathrm{dgm}(\mathbb{V})$ *is the union of the points of the diagonal Δ of $\mathbb{R}^2$, counted with infinite multiplicity, and a multi-set above the diagonal in $\mathbb{R}^2$ that is locally finite. Here, by locally finite we mean that for any rectangle R with sides parallel to the coordinate axes that does not intersect Δ, the number of points of* $\mathrm{dgm}(\mathbb{V})$*, counted with multiplicity, contained in R is finite. Also, the part of the diagram made of the points with infinite second coordinate is called the essential part of the diagram.*

The construction of persistence diagrams of q-tame modules is beyond the scope of this paper but it gives rise to the same notion as in the case of decomposable modules. It can be done either by following the algebraic approach based upon the decomposability properties of modules, or by adopting a measure theoretic approach that allows to define diagrams as integer valued measures on a space of rectangles in the plane. We refer the reader to Chazal et al. (2016a) for more informations.

Although persistence modules encountered in practice are decomposable, the general framework of q-tame persistence module plays a fundamental role in the mathematical and statistical analysis of persistent homology. In particular, it is needed to ensure the existence of limit diagrams when convergence properties are studied - see Section 6.

A filtration $\mathrm{Filt} = (F_r)_{r \in T}$ of a simplicial complex or of a topological space is said to be tame if for any integer k, the persistence module $(H_k(F_r) \mid r \in T)$ is q-tame. Notice that the filtrations of finite simplicial complexes are always tame. As a consequence, for any integer k a persistence diagram denoted $\mathrm{dgm}_k(\mathrm{Filt})$ is associated to the filtration Filt. When k is not explicitly specified and when there is no ambiguity, it is usual to drop the index k in the notation and to talk about "the" persistence diagram $\mathrm{dgm}(\mathrm{Filt})$ of the filtration Filt. This notation has to be understood as "$\mathrm{dgm}_k(\mathrm{Filt})$ for some k".

5.4 Persistence landscapes

The persistence landscape introduced in Bubenik (2015) is an alternative representation of persistence diagrams. This approach aims at representing the topological information encoded in persistence diagrams as elements of an Hilbert space, for which statistical learning methods can be directly applied. The persistence landscape is a collection of continuous, piecewise linear functions $\lambda \colon \mathbb{N} \times \mathbb{R} \to \mathbb{R}$ that summarizes a persistence diagram dgm.

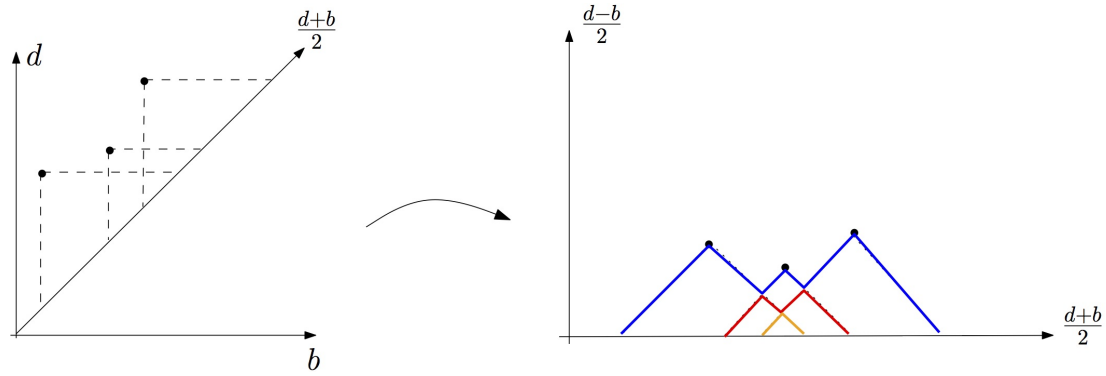

Figure 11: An example of persistence landscape (right) associated to a persistence diagram (left). The first landscape is in blue, the second one in red and the last one in orange. All the other landscapes are zero.

A birth-death pair $p = (b, d) \in \mathrm{dgm}$ is transformed into the point $\left(\frac{b+d}{2}, \frac{d-b}{2}\right)$, see Figure 11. Remember that the points with infinite persistence have been simply discarded in this definition. The landscape is then defined by considering the set of functions created by tenting the features of the rotated persistence diagram as follows:

$$\Lambda_p(t) = \begin{cases} t - b & t \in [b, \frac{b+d}{2}] \\ d - t & t \in (\frac{b+d}{2}, d] \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

The persistence landscape λ_{dgm} of dgm is a summary of the arrangement of piecewise linear curves obtained by overlaying the graphs of the functions $\{\Lambda_p\}_{p \in \mathrm{dgm}}$. Formally, the persistence landscape of dgm is the collection of functions

$$\lambda_{\mathrm{dgm}}(k, t) = \underset{r \in \mathrm{dgm}}{\mathrm{kmax}}\ \Lambda_r(t), \quad t \in [0, T], k \in \mathbb{N}, \tag{4}$$

where kmax is the kth largest value in the set; in particular, 1max is the usual maximum function. Given $k \in \mathbb{N}$, the function $\lambda_{\mathrm{dgm}}(k, .) : \mathbb{R} \to \mathbb{R}$ is called the *k-th landscape* of dgm. It is not difficult to see that the map that associate to each persistence diagram its corresponding landscape is injective. In other words, formally no information is lost when a persistence diagram is represented through its persistence landscape.

The advantage of the persistence landscape representation is two-fold. First, persistence diagrams are mapped as elements of a functional space, opening the door to the use of a broad

variety of statistical and data analysis tools for further processing of topological features - see, e.g. Bubenik (2015); Chazal et al. (2015b) and Section 6.3.1. Second, and fundamental from a theoretical perspective, the persistence landscapes share the same stability properties as persistence diagrams - see Section 5.7.

5.5 Linear representations of persistence homology

A persistence diagram without its essential part can be represented as a discrete measure on $\Delta^+ = \{p = (b, d),\ b < d < \infty\}$. With a slight abuse of notation, we can write :

$$\mathrm{dgm} = \sum_{p \in \mathrm{dgm}} \delta_p,$$

where the features are counted with multiplicity and where $\delta_{(b,d)}$ denotes the Dirac measure in $p = (b, d)$. Most of the persistence-based descriptors that have been proposed to analyze persistence can be expressed as linear transformations of the persistence diagram, seen as a point process

$$\Psi(\mathrm{dgm}) = \sum_{p \in \mathrm{dgm}} f(p),$$

for some function f defined on Δ and taking values in a Banach space.

In most cases, we want these transformations to apply independently at each homological dimension. For $k \in \mathbb{N}$ a given homological dimension we then consider some linear transformation of the persistence diagram restricted to the topological features of dimension k:

$$\Psi_k(\mathrm{dgm}_k) = \sum_{p \in \mathrm{dgm}_k} f_k(p), \tag{5}$$

where dgm_k is the persistence diagram of the topological features of dimension k and where f_k is defined on Δ and takes values in a Banach space.

Betti curve. The simplest way to represent persistence homology is the Betti function or Betti curve. The Betti curve of homological dimension k is defined as

$$\beta_k(t) = \sum_{(b,d) \in \mathrm{dgm}} w(b, d) \mathbf{1}_{t \in [b,d]}$$

where w is a weight function defined on Δ. In other words, the Betti curve is the number of barcodes at time m. This descriptor is a linear representation of persistence homology by taking f in (5) such that $f(b, d)(t) = w(b, d)\mathbf{1}_{t \in [b,d]}$. A typical choice for the weigh function is a increasing function of the persistence $w(b, d) = \tilde{w}(d - b)$ where $\tilde{w}$ is an increasing function defined on $\mathbb{R}^+$. One of the first applications of Betti curves can be found in Umeda (2017).

Persistence surface. Persistence surface (also called persistence images) is obtained by making the convolution of a diagram with a kernel. It has been introduced in Adams et al. (2017). For $K : \mathbb{R}^2 \to \mathbb{R}$ a kernel and H a 2×2 bandwidth matrix (e.g. a symmetric positive definite matrix), let for $u \in \mathbb{R}^2$

$$K_H(u) = \det(H)^{-1/2} K(H^{-1/2} u).$$

Let $w : \mathbb{R}^2 \to \mathbb{R}_+$ a weight function defined on Δ. One defines the persistence surface of homological dimension k associated to a diagram dgm, with kernel K and bandwidth matrix H by:

$$\forall u \in \mathbb{R}^2,\ \rho_k(\mathrm{dgm})(u) = \sum_{p \in \mathrm{dgm}_k} w(r) K_H(u - p).$$

The persistence surface is obviously a linear representation of persistence homology. Typical weigh functions are increasing functions of the persistence.